Adiabat transform

Change of variables applied before the usual (grouped) mean/std normalization. Goal: inputs/outputs that shift less between climates and datasets (piControl → 1pct/abrupt-CO2, ERA5 ↔ C96/CM4/X-SHiELD).

Level indexing: $k = 0$ is the top (stratosphere), $k = 7$ is the lowest level.

1. Reference parcel (per column, per timestep)

Inputs: T_7, q_7 (`specific_total_water_7`), p_s (`PRESsfc`). Level pressures: $p_k = a_k + b_k\, p_s$.

Saturation humidity (Bolton e_s, hPa):

$$e_s(T) = 6.112\, \exp\!\left(\frac{17.67\,(T - 273.15)}{T - 29.65}\right), \qquad q_{\text{sat}}(T, p) = \frac{\varepsilon\, e_s(T)}{p - (1 - \varepsilon)\, e_s(T)}$$

Lift a parcel from (T_7, q_7, p_7) upward:

1. **Dry ascent** (humidity fixed at q_7, below saturation):

$$T_p(p) = T_7 \left(\frac{p}{p_7}\right)^{R_d/c_p}$$

   until $q_{\text{sat}}(T_p, p) = q_7$ (saturation level). Skip if $q_7 \geq q_{\text{sat}}(T_7, p_7)$.
2. **Moist ascent** (saturated, excess vapor condenses and releases heat):

$$\frac{dT_p}{dp} = \frac{1}{p}\,\frac{R_d T_p + L_v q_s}{c_p + \dfrac{\varepsilon L_v^2 q_s}{R_d T_p^2}}, \qquad q_s = q_{\text{sat}}(T_p, p)$$

3. Record $T_p(p_k)$ for $k = 1, \ldots, 6$.

2. Forward transform

Variable	Transformed	Levels
Temperature	$D_k = T_k - T_p(p_k)$	$k = 1..6$
Humidity	$H_k = q_k / q_{\text{sat}}(T_k, p_k)$	$k = 0..6$
Everything else	unchanged	T_0, T_7, q_7, winds, surface, forcings

T_7, q_7, p_s stay raw because they define the parcel. H_k can exceed 1 in clouds (q includes condensate); that's fine.

Then normalize as today, per group g: $\hat{D}_k = (D_k - \mu_{g,k})/\sigma_{g,k}$, same for H_k.

3. Inverse (after network output)

1. Rebuild $T_p(p_k)$ from predicted T_7, q_7, p_s.
2. $T_k = D_k + T_p(p_k)$.

3. $q_k = H_k \, q_{\text{sat}}(T_k, p_k)$ (needs step 2 first).

Exact inverse; loss computed on physical T_k, q_k (keeps metrics comparable).

4. Implementation notes

- Differentiable torch op in `fme/core`, applied before normalizer / after denormalizer; config saved with stepper for checkpoint loading.
- Moist ascent: fixed RK substeps in $\ln p$ between levels; saturation level via Newton or Bolton LCL formula.
- Liquid-only latent heat to start (no ice).
- Code change goes in a PR off `main`, not the FM branch.

5. Plan

1. **Offline check (no training).** Per group, compute D_k, H_k. Compare gap $|\mu_a - \mu_b|/\sigma$ for raw vs transformed variables: piControl vs 1pct, ERA5 vs other groups. Tropics and extratropics separately. Stop if gaps don't clearly shrink.
2. **Decide levels.** Include/exclude T_1, H_0 based on step 1.
3. **One training arm** vs current baseline, only this change. Judge on abrupt-CO2 / 1pct generalization and cross-dataset skill, not 5-day RMSE.

Risks

- Extratropics/poles not near moist adiabat → large, noisy D_k.
- Error in predicted T_7, q_7 propagates to all levels on inverse.
- Expected gain modest; mostly climate generalization, little weather skill.